

5.1 Chemical Energetics (A Level Only)

Question Paper

Course	CIE A Level Chemistry
Section	5. Physical Chemistry (A Level Only)
Topic	5.1 Chemical Energetics (A Level Only)
Difficulty	Easy

Time allowed: 40

Score: /31

Percentage: /100

Question 1a

This question is about the formation of calcium oxide.

Explain the meaning of the term lattice energy.

[3 marks]

Question 1b

The Born-Haber cycle in Fig. 1.1 can be used to determine the lattice enthalpy of calcium oxide.

Steps **A-G** includes the values for enthalpy changes.

Complete the Born-Haber cycle by adding the species present on the two dotted lines.

Include state symbols.

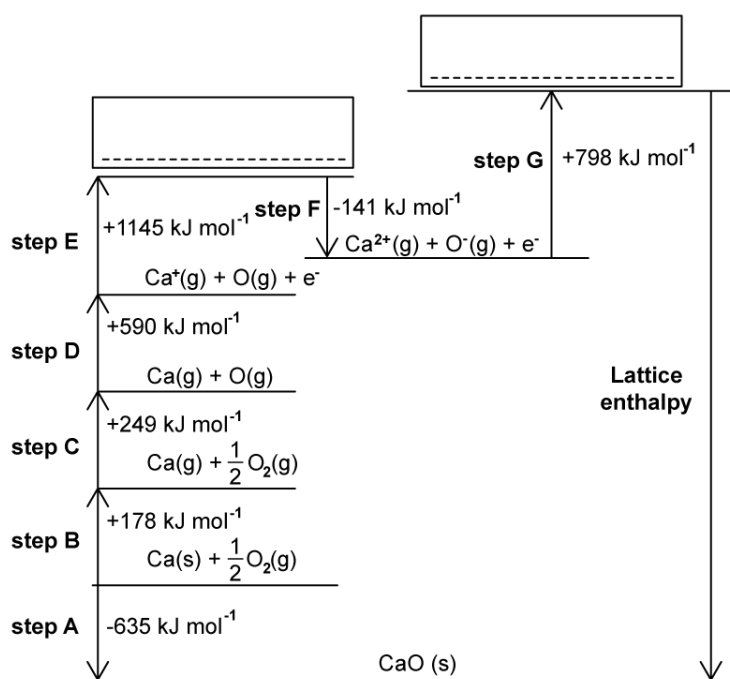


Fig. 1.1

[2 marks]

Question 1c

Name the enthalpy changes for the following steps in the Born-Haber cycle.

- Step B
- Step D
- Step F

[3 marks]

Question 1d

Step C represents the enthalpy change of atomisation of oxygen.

Why is the enthalpy change of atomisation always a positive value?

[2 marks]

Question 2a

This question is about lattice enthalpy.

Write one equation to represent the following changes:

- Atomisation of sodium
- Second ionisation energy of magnesium
- First electron affinity of chlorine

[3 marks]

Question 2b

The Born-Harber cycle for the formation of potassium fluoride is shown in Fig. 2.1.

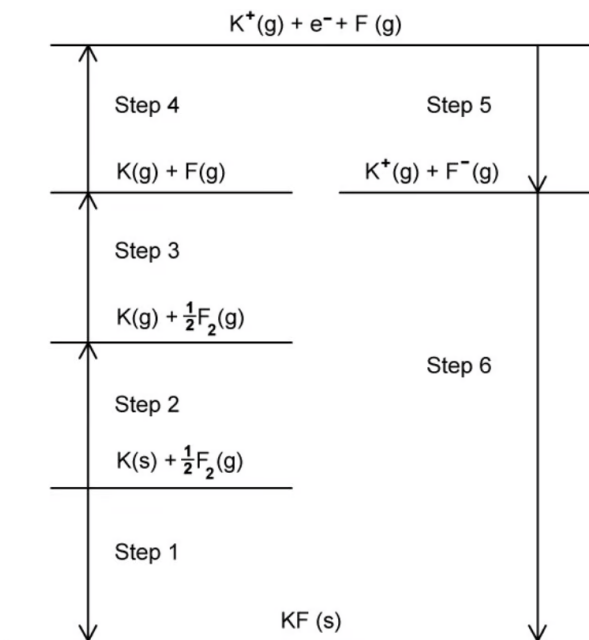


Fig. 2.1

Complete Table 2.1 by naming the enthalpy changes associated with the identified steps.

Table 2.1

Step	Name of the Enthalpy Change
1	
2	Atomisation of potassium
3	
4	First ionisation energy of potassium
5	
6	Lattice enthalpy of formation

[3 marks]

Question 2c

The enthalpies of lattice formation for potassium fluoride and caesium fluoride are -830 kJ mol^{-1} and -730 kJ mol^{-1} respectively.

Explain why the enthalpy of lattice formation is more exothermic for potassium fluoride.

[3 marks]**Question 2d**

Use the data in Table 2.2 to calculate the enthalpy of solution of potassium fluoride.

Table 2.2

Enthalpy change	Enthalpy change (kJ mol^{-1})
$\Delta H^\theta_{\text{latt}} \text{KF}$	-830
$\Delta H^\theta_{\text{hyd}} \text{K}^+$	-351
$\Delta H^\theta_{\text{hyd}} \text{F}^-$	-504

$$\Delta H^\theta_{\text{sol}} \text{KF} = \dots\dots\dots \text{kJ mol}^{-1}$$

[3 marks]

Question 3a

This question is about enthalpy changes and calculations.

Table 3.1 shows the enthalpy data for the formation of sodium chloride.

Table 3.1

Enthalpy change	Enthalpy change	Enthalpy change / kJ mol^{-1}
$\Delta H^\theta_f \text{NaCl}$	$\text{Na (s)} + \frac{1}{2}\text{Cl}_2\text{(g)} \rightarrow \text{NaCl(s)}$	-411
$\Delta H^\theta_{\text{at}} \text{Cl}$	$\frac{1}{2}\text{Cl}_2\text{(g)} \rightarrow \text{Cl(g)}$	121
$\Delta H^\theta_{\text{at}} \text{Na}$	$\text{Na (s)} \rightarrow \text{Na (g)}$	108
$E_{\text{ea}} \text{Cl}$	$\text{Cl(g)} \rightarrow \text{Cl}^-\text{(g)}$	-349
$\Delta H^\theta_{\text{ie}} \text{Na}$	$\text{Na (g)} \rightarrow \text{Na}^+\text{(g)}$	496
$\Delta H^\theta_{\text{latt}} \text{NaCl}$	$\text{Na}^+\text{(g)} + \text{Cl}^-\text{(g)} \rightarrow \text{NaCl(s)}$	To be calculated

Calculate the enthalpy of lattice formation of sodium chloride.

$$\Delta H^\theta_{\text{latt}} \text{NaCl} = \dots\dots\dots \text{kJ mol}^{-1}$$

[3 marks]

Question 3b

State the definition of electron affinity.

[3 marks]

Question 3c

A section of the Born-Haber cycle for the formation of magnesium oxide is shown in Fig. 3.1.

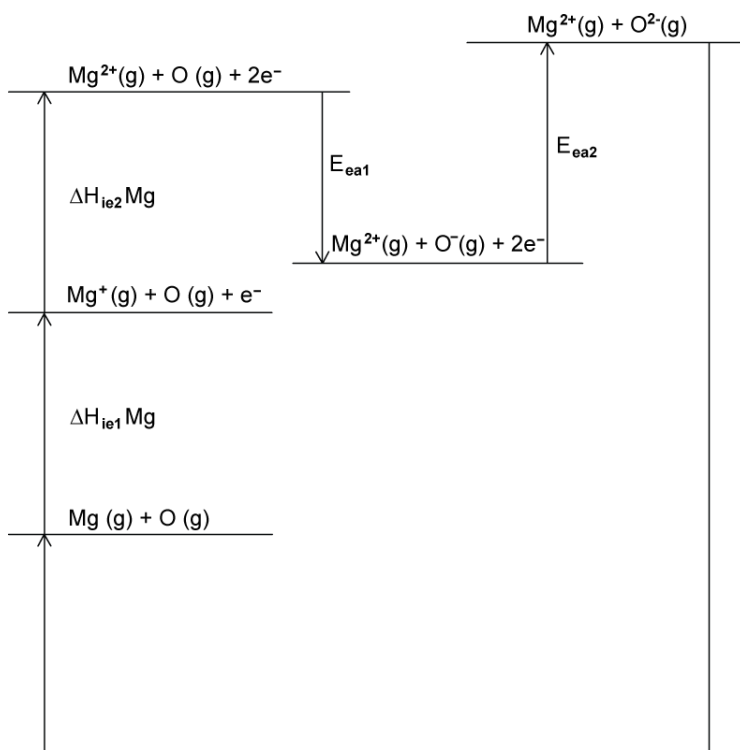


Fig. 3.1

The first electron affinity of oxygen has a negative value so the arrow points downwards. The second electron affinity of oxygen has a positive value so the arrow points upwards.

Explain why the arrows point in different directions.

[3 marks]

Model Answers: Easy

1a

a) Lattice energy is:

- The formation of 1 mole of an ionic lattice; [1 mark]
- From gaseous ions; [1 mark]
- Under standard conditions; [1 mark]

[Total: 3 marks]

- There are many thermodynamic definitions that you have to know
 - They are typically worth 2 or 3 marks, so they are definitely worth knowing

1b

b) The missing information is:

- (Step **E**) $\text{Ca}^{2+}(\text{g}) + \text{O}(\text{g}) + 2\text{e}^{-}$; [1 mark]
- (Step **G**) $\text{Ca}^{2+}(\text{g}) + \text{O}^{2-}(\text{g})$; [1 mark]

[Total: 2 marks]

- Step **E** begins with $\text{Ca}^{+}(\text{g}) + \text{O}(\text{g}) + 2\text{e}^{-}$
 - The calcium ion needs to lose another electron to form a 2+ ion (second ionisation energy) so $\text{Ca}^{2+}(\text{g}) + \text{O}(\text{g}) + 2\text{e}^{-}$ is the missing species
- Step **G** represents the second electron affinity of oxygen
 - This is the enthalpy change when one mole of gaseous ions with a single negative charge forms one mole of gaseous ions with a double negative charge
 - As shown in the Born Haber cycle at Step **G** $\text{Ca}^{2+}(\text{g}) + \text{O}^{-}(\text{g}) + \text{e}^{-}$ becomes $\text{Ca}^{2+}(\text{g}) + \text{O}^{2-}(\text{g})$
 - Examiners love asking about the second electron affinity so make sure you know it!

1c

c) The correct enthalpy changes are:

- (B) enthalpy of atomisation; [1 mark]
- (D) first ionisation energy of calcium; [1 mark]
- (F) first electron affinity of oxygen; [1 mark]

[Total: 3 marks]

- The enthalpy of atomisation is the enthalpy change for the formation of one mole of gaseous atoms from the element its standard state,
 - Ca(s) becomes Ca(g)
- The first ionisation energy is the enthalpy change when one mole of electrons is removed from one mole of gaseous atoms to form one mole of gaseous ions with a single positive charge
 - Ca(g) becomes $\text{Ca}^+(\text{g})$
- The first electron affinity is the enthalpy change when one mole of gaseous atoms forms one mole of negative ions with a single negative charge
 - O(g) becomes $\text{O}^-(\text{g})$

1d

d) The enthalpy of atomisation is always positive because:

- Energy is taken in (to break the bonds); [1 mark]
- (So) the reaction is endothermic; [1 mark]

[Total: 2 marks]

- The enthalpy of atomisation is the enthalpy change when 1 mole of gaseous atoms is formed from its element under standard conditions e.g Na(s) becomes Na(g)
- The reaction is endothermic as energy is taken in for this process to break the bonds
- **Remember:** Endothermic reactions always have positive enthalpy changes and exothermic have negative

2a

a) The equations are:

- $\text{Na (s)} \rightarrow \text{Na (g)}$; [1 mark]
- $\text{Mg}^+ \text{(g)} \rightarrow \text{Mg}^{2+} \text{(g)} + \text{e}^-$; [1 mark]
- $\text{Cl (g)} + \text{e}^- \rightarrow \text{Cl}^- \text{(g)}$; [1 mark]

[Total: 3 marks]

- Make sure you learn the definitions of these terms as well as being able to write equations for them

2b

b) The completed Table 2.1 is:

Step	Name of the Enthalpy Change
1	Enthalpy change of formation; [1 mark]
2	Atomisation of potassium
3	Atomisation of fluorine; [1 mark]
4	First ionisation energy of potassium
5	First electron affinity of fluorine; [1 mark]
6	Lattice enthalpy of formation

- Each correct step scores 1 mark

[Total: 3 marks]

- Labelling steps in a Born-Haber cycle is an important skill so again, be careful to take time learning the changes required

2c

c) The enthalpy of lattice formation for potassium fluoride is more exothermic than that of caesium fluoride because:

- The potassium ion is smaller than the caesium ion

OR

The potassium ion has fewer electron shells; [1 mark]

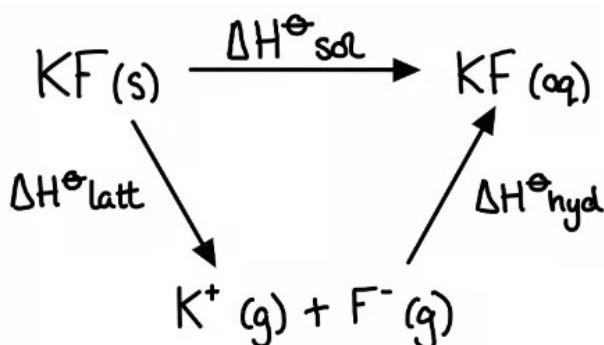
- Therefore the distance between the potassium ion and fluoride ion is shorter; [1 mark]
- Therefore the potassium - fluoride / KF bonding is stronger; [1 mark]

[Total: 3 marks]

- When comparing lattice enthalpies there are three things to take into account:
 - The **geometry** of the ionic solid
 - The **charge** on the ions
 - The **distance** between the ions
- **Remember:** The size of the ions increases down Group 1 due to the increased number of electron shells
- The shorter the inter-ionic distance, the stronger the attraction with another ion

2d

d) To calculate the enthalpy change of solution for potassium fluoride:



- $\Delta H^\ominus_{\text{sol}} = \Sigma \Delta H^\ominus_{\text{hyd}} - \Delta H^\ominus_{\text{latt}}$

OR

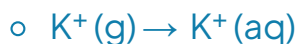
Correctly constructed cycle; [1 mark]

- $\Delta H^\ominus_{\text{sol}} = (-351 + -504 - (-830))$; [1 mark]
- $\Delta H^\ominus_{\text{sol}} = -25 \text{ kJ mol}^{-1}$; [1 mark]

[Total: 3 marks]

- The enthalpy change of solution ($\Delta H^\ominus_{\text{sol}}$) is the enthalpy change when one mole of an ionic substance dissolves in water to give a solution of infinite dilution
 - $\text{KF (s)} \rightarrow \text{KF (aq)}$
- The enthalpy of hydration ($\Delta H^\ominus_{\text{hyd}}$) is the enthalpy change when one mole of

gaseous ions dissolve in water to give a solution of infinite dilution



- $\Delta H^\theta_{sol} = \Sigma \Delta H^\theta_{hyd} - \Delta H^\theta_{latt}$ is the correct equation to use to calculate the enthalpy change of solution
- **Careful:** Make sure that you include the sum (Σ) of hydration enthalpies for each ion in the ionic compound

3a

a) To calculate the enthalpy of lattice formation for sodium chloride:

- $\Delta H^\theta_f(\text{NaCl}) = \Delta H^\theta_{at}(\text{Na}) + \Delta H^\theta_{at}(\text{Cl}) + E_{ea}(\text{Cl}) + \Delta H^\theta_{ie}(\text{Na}) + \Delta H^\theta_{latt}(\text{NaCl})$

OR

$$\Delta H^\theta_{latt}(\text{NaCl}) = -E_{ea}(\text{Cl}) - \Delta H^\theta_{ie}(\text{Na}) - \Delta H^\theta_{at}(\text{Cl}) - \Delta H^\theta_{at}(\text{Na}) + \Delta H^\theta_f(\text{NaCl}); [1 \text{ mark}]$$

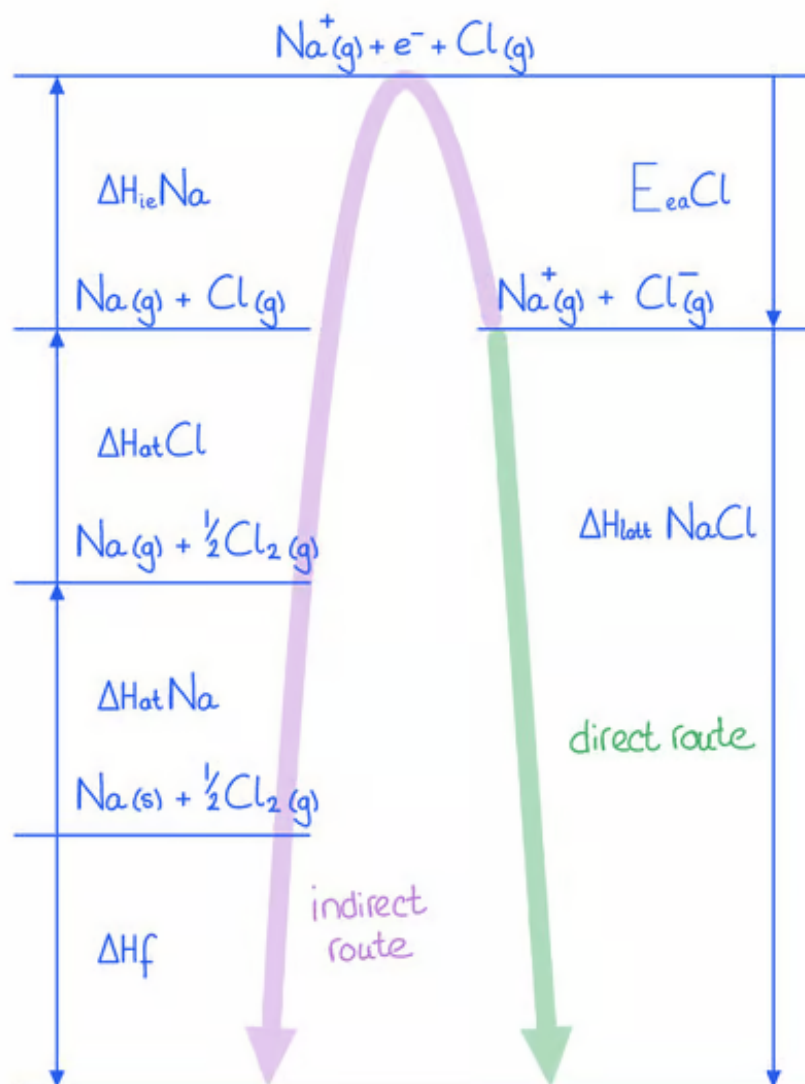
- $\Delta H^\theta_{latt}(\text{NaCl}) = -(-349) - 496 - 121 - 108 - 411; [1 \text{ mark}]$

- $\Delta H^\theta_{latt}(\text{NaCl}) = -787 \text{ (kJ mol}^{-1}\text{)}; [1 \text{ mark}]$

[Total: 3 marks]

- When calculating the enthalpy of lattice formation, you can draw a Born-Haber cycle to help you
- The direct route is the same as the indirect route

- The direct route is the same as the indirect route



3b

b) Electron affinity is:

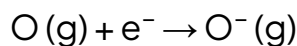
Alternative 1

- The electron affinity of an element is the energy change when one mole of

the electron affinity of an element is the energy change when one mole of electrons is gained; [1 mark]

- By one mole of gaseous atoms of an element to form one mole of gaseous 1- ions; [1 mark]
- Under standard conditions; [1 mark]

Alternative 2



- Correct equation with state symbols; [2 marks]
- Under standard conditions; [1 mark]

[Total: 3 marks]

- This is another definition which is key to learn!
- Also, make sure you can represent this with an equation to illustrate the definition

3c

c) The reason why the arrows point in opposite directions

- The first electron affinity of oxygen is exothermic; [1 mark]
- The second electron affinity of oxygen is endothermic; [1 mark]
- Because the repulsion between electron and negative ion must be overcome (requiring a large amount of energy); [1 mark]

[Total: 3 marks]

- In a Born-Haber cycle, if an arrow points upwards, the energy change is positive therefore is an endothermic process
- if an arrow points downwards, the energy change is negative therefore is an exothermic process
- The first electron affinity of oxygen is exothermic
 - $\text{O}(\text{g}) + \text{e}^{-} \rightarrow \text{O}^{-}(\text{g})$ $E_{\text{ea}1} = \text{exothermic}$
- As energy is released when electrons are attracted to the atoms
- The **second electron affinity** of an element can be **endothermic**
 - $\text{O}^{-}(\text{g}) + \text{e}^{-} \rightarrow \text{O}^{2-}(\text{g})$ $E_{\text{ea}2} = \text{endothermic}$

